Guardrail AI – Behavioural Co-Pilot for Retail Traders

Concept Report

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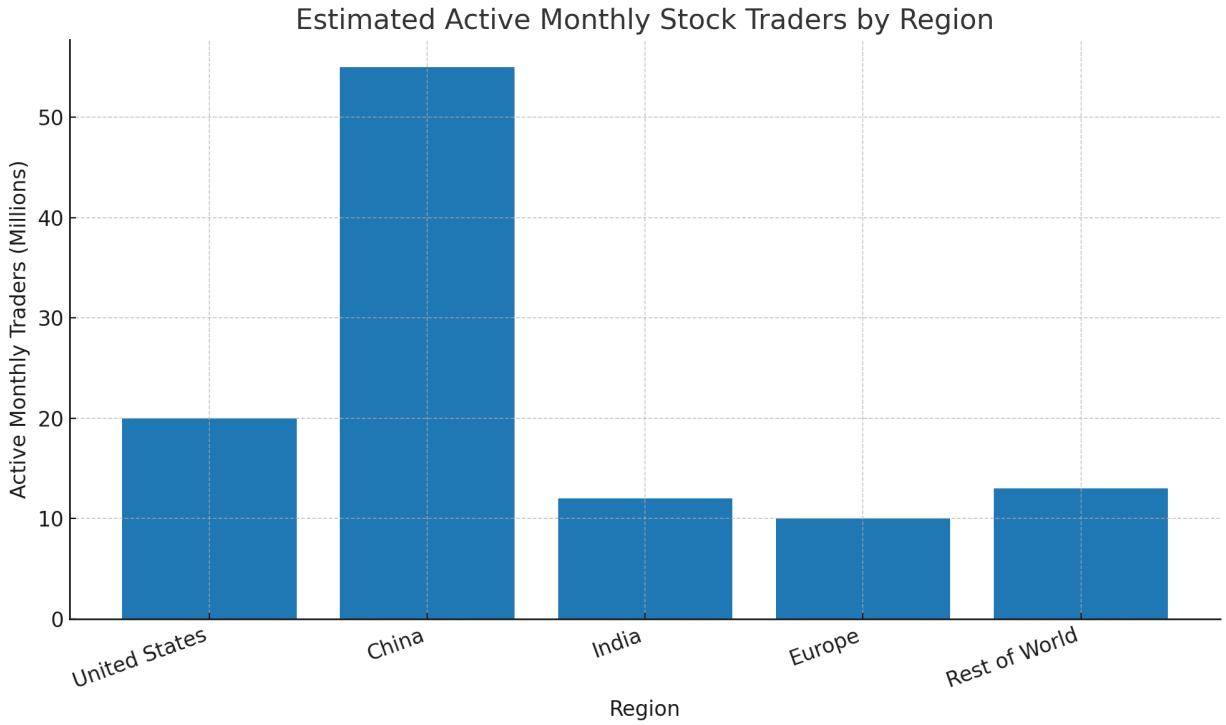
Executive Summary

Guardrail AI is an independent, broker-agnostic AI agent that helps retail traders avoid costly behavioural and risk mistakes. Instead of providing stock tips, Guardrail operates at the moment of execution: it frames decisions with balanced upside/downside projections, enforces user-defined guardrails (position sizing, loss caps, cool-downs), and auto-journals trades for structured weekly debriefs. This approach tackles the root causes of underperformance—psychology, risk, and discipline—within an EU-compliant perimeter (no investment advice).

Target users are the ~100M+ people worldwide who place at least one stock trade per month, including an estimated 8–12M in Europe. The product launches as a companion app and browser overlay (fast to market), then deepens distribution via broker SDK partnerships. Primary revenue streams include Pro subscriptions and B2B SDK licensing.

Chapter 1. Background

Retail participation in public markets is significant and growing. While hundreds of millions of people own stocks, a smaller but still vast segment executes monthly trades. Guardrail focuses on this active, high-intent audience. The chart below illustrates the estimated distribution of monthly active traders by region (illustrative for sizing purposes).

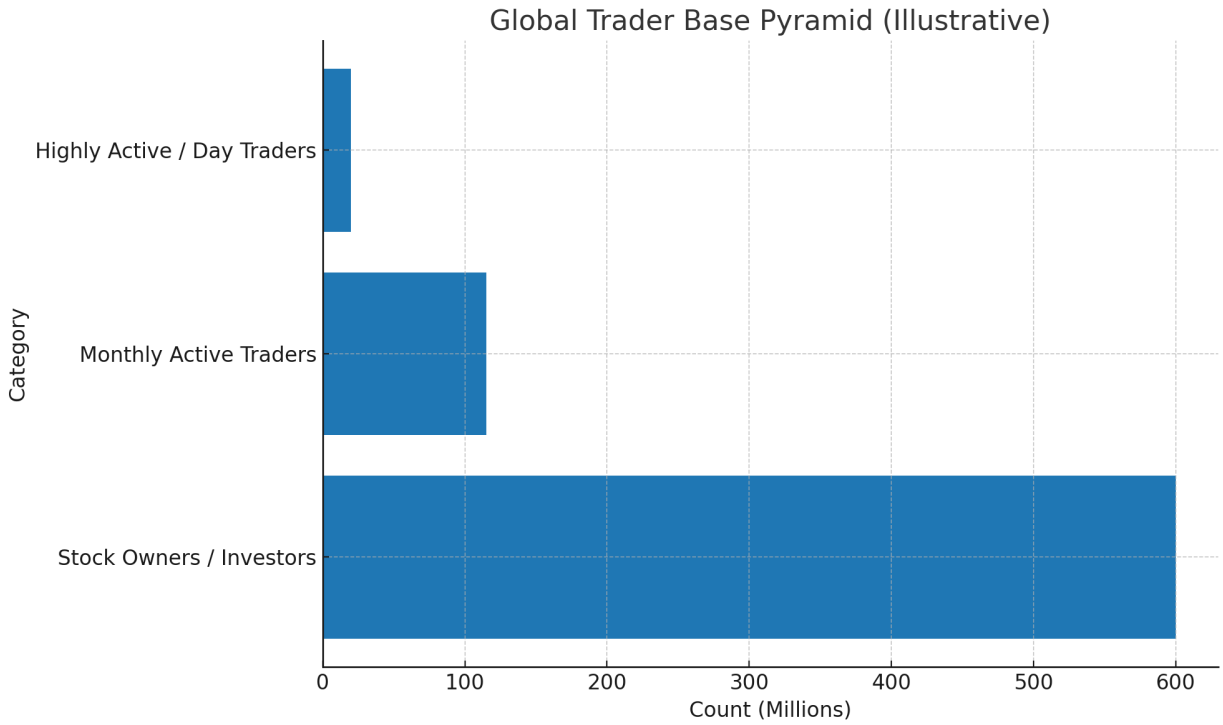


Estimated Active Monthly Stock Traders by Region (Millions):

Region	Active Monthly Traders (M)

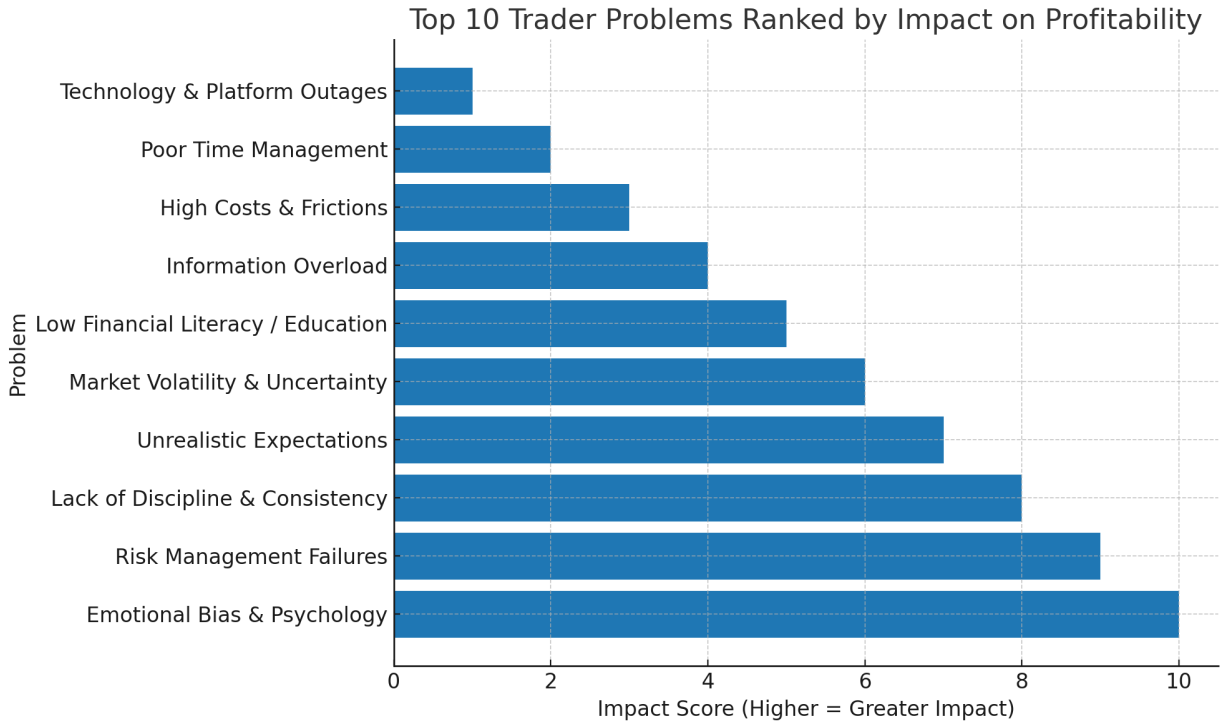
United States	20
China	55
India	12
Europe	10
Rest of World	13

Trader base segmentation matters for positioning and pricing. Most stock owners are passive; only a subset trades monthly; a smaller subset trades daily. Guardrail is built for the two most engaged cohorts.



Chapter 2. Problem

Active retail traders consistently face behavioural and risk challenges that erode profitability. The chart below ranks the issues by their impact on outcomes. Most are psychological or process-driven rather than data-availability problems.



Top pain points (summary):

- Emotional bias drives poor entries/exits and revenge trading.
- Risk management failures (oversizing, leverage misuse, no hard loss caps).
- Lack of discipline/consistency (strategy hopping).
- Unrealistic expectations (chasing lottery-like payoffs).
- Volatility shocks and event risk hit unhedged positions.
- Low financial literacy and limited risk foundations.
- Information overload vs. signal; limited attention.
- Costs & frictions (spreads, slippage, financing).
- Time constraints for part-time traders.
- Platform outages (episodic but severe).

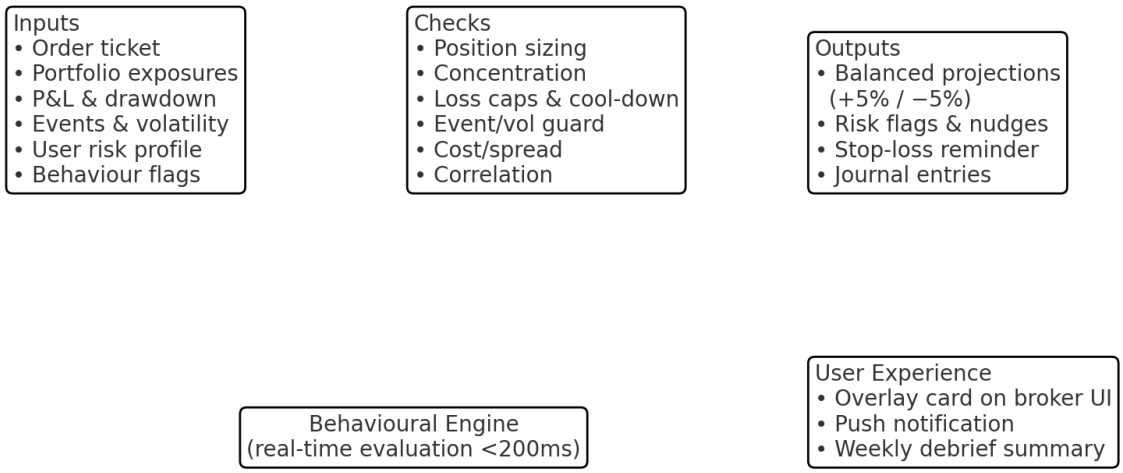
Chapter 3. Solution

Guardrail AI intervenes at the moment of intent and execution. It frames decisions with both upside and downside outcomes, checks trades against user-defined rules, and guides behaviour with real-time nudges. It then journals decisions and outcomes to make reflection habitual.

Core capabilities:

- Real-time guardrail rule engine at order time (sub-200ms checks).
- Balanced positive/negative projections tied to the order (e.g., “+5% gain would add +2.3% to the portfolio; −5% loss would cut it by −12% and breach your risk cap”).
- Behavioural nudges (reduce size, add stop, cool-down after consecutive losses).
- Auto-journal + weekly debriefs (adherence, cost drag, lessons).
- Privacy-first design (EU data residency; minimal collection; on-device inference for sensitive prompts).
- Optional accountability pods for lightweight peer review.

Guardrail Rule Engine – simplified flow:



Example user notification (pre-trade scenario card):

Pre-Trade Scenario Card

Order: BUY 40 shares of TSLA @ €200

If price rises +5%:
- Position P&L: +€400
- Portfolio impact: +2.3%

If price falls −5%:
- Position P&L: −€400
- Portfolio impact: −12% (breaches risk cap)

Suggestion: Reduce size to 25 shares or add stop-loss @ €190

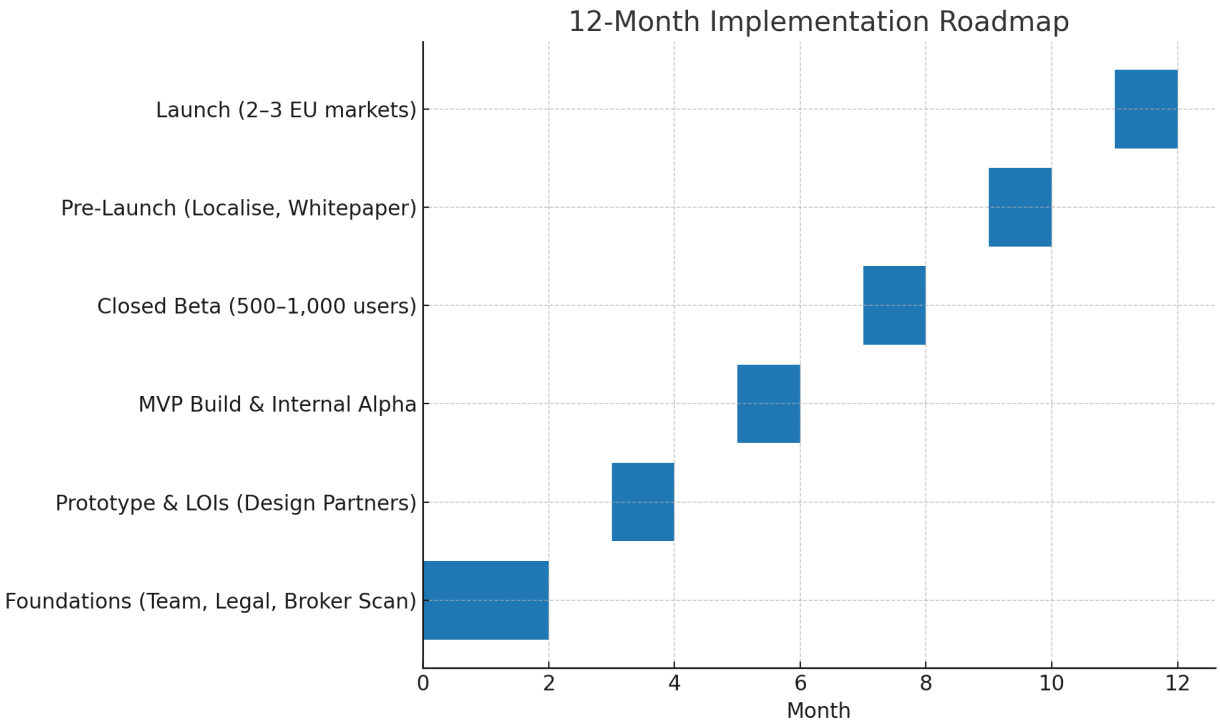
Chapter 4. Business Model & Market Entry

Monetisation streams:

- Freemium (core guardrails, journaling)
- Pro (€12/mo): advanced rules, scenarios, pods
- SDK/API licensing to brokers (per-MAU fees)

Go-to-market (phased):

- Phase 1 – Companion App (fast to market; read-only + nudges)
- Phase 2 – Browser Overlay (real-time prompts on web broker UIs)
- Phase 3 – Broker SDK (embedded pre-trade checks; retention feature for platforms)



Chapter 5. Competitor Analysis

No single product today combines real-time guardrails, portfolio-aware projections, and behavioural nudges at order time across brokers. The table below summarises adjacent solutions and key gaps.

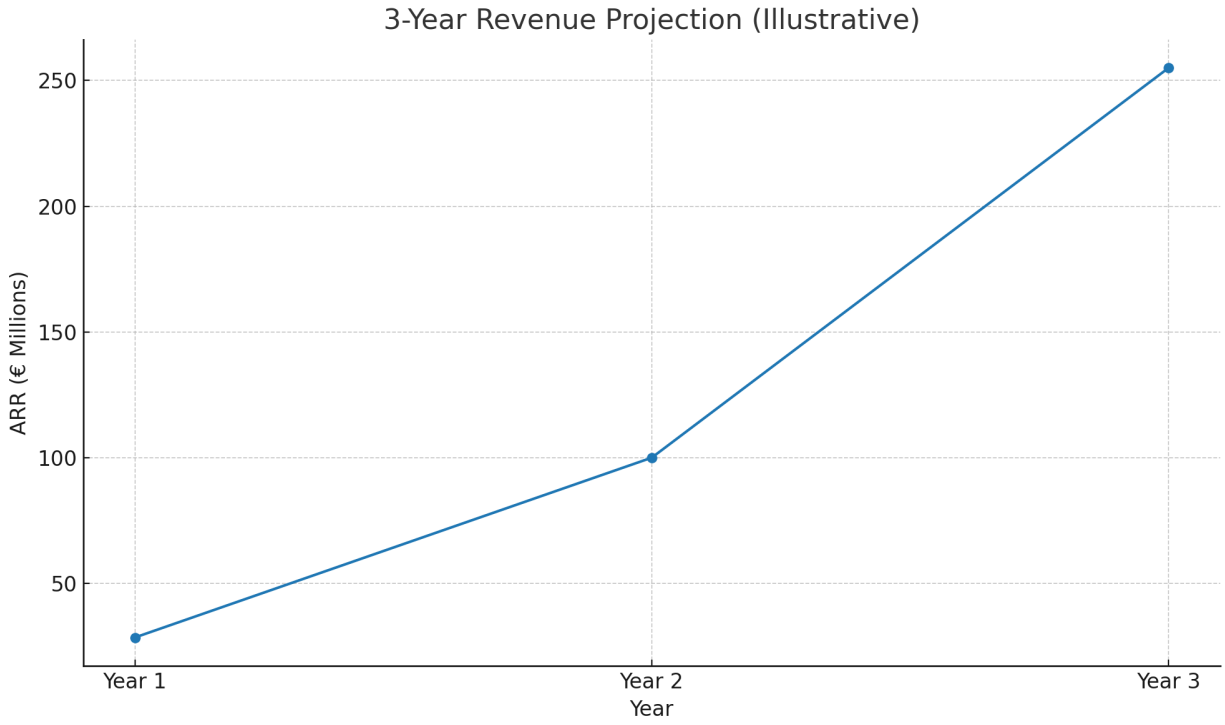
Product / Category	Strengths	Gaps vs. Guardrail
TraderSync / Tradervue (journals)	Strong post-trade analytics	No real-time order-time guardrails or balanced projections; platform-agnostic journaling only

Capitalise.ai (no-code automation)	Strategy automation via NL; some partner brokers	Focus on executing ideas; heavier regulatory footprint; not behaviour-first
cTrader Risk/Reward Tool	Visual risk-reward on chart	Platform-bound; lacks portfolio context; no proactive nudges
OANDA VaR Tool	Illustrates risk ranges (educational)	Not tied to execution; not portfolio-aware
Broker Alerts (e.g., Schwab)	Event/performance alerts	After-the-fact; not pre-trade scenario framing

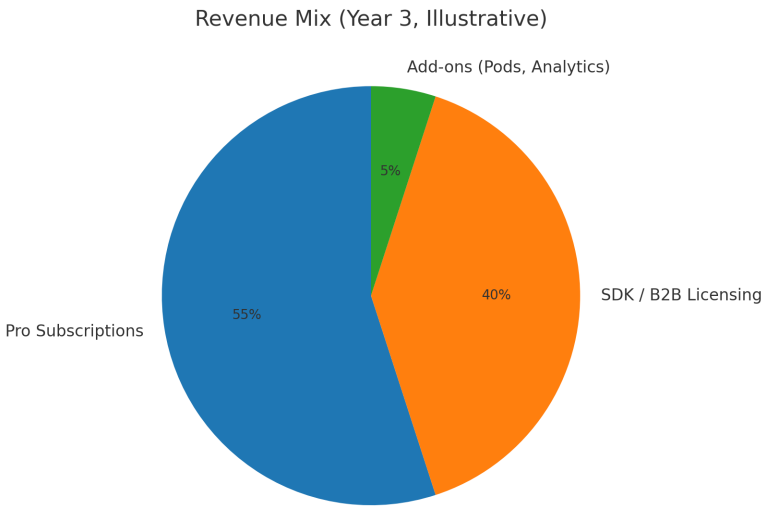
Guardrail's differentiation: real-time, portfolio-aware, positive/negative projections embedded in execution flow, coupled with behavioural guardrails and journaling. Broker-agnostic, EU-first compliance posture.

Chapter 6. Revenue Projections & Strategy

Revenue assumes dual B2C (Pro subscriptions) and B2B2C (SDK licensing) streams. Values below are illustrative and depend on broker partnerships and conversion rates.



Year 3 illustrative revenue mix:



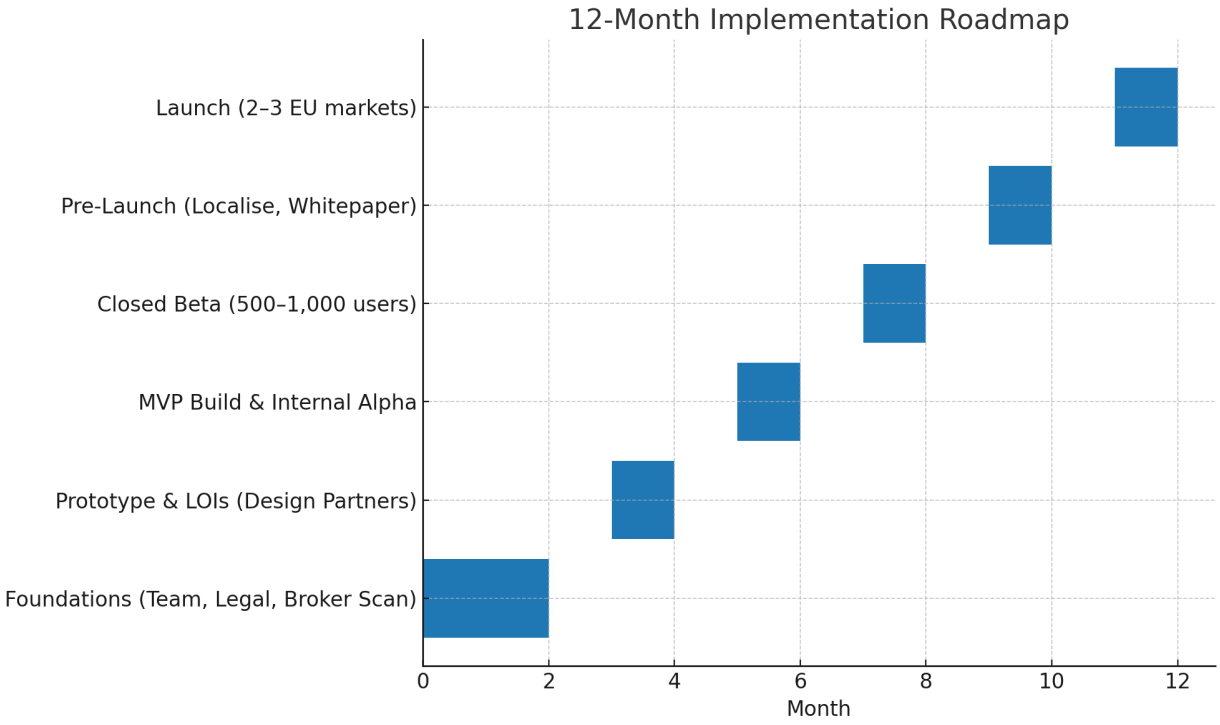
Chapter 7. Defensibility

Why incumbents won't easily replicate Guardrail:

- Incentive misalignment: many brokers monetise order flow/volume; Guardrail reduces overtrading.
- Regulatory risk: embedding strong guardrails within brokers can raise scrutiny; an independent layer avoids advice perimeters.
- Roadmap constraints: brokers prioritise growth features (payments, crypto, ETFs, AI signals).
- Cross-broker scope: traders use multiple platforms; Guardrail unifies risk view across accounts.
- Behavioural dataset moat: longitudinal, labelled actions→outcomes enable better nudges over time.
- Trust & independence: users prefer neutral tools (not controlled by brokers).
- EU first-mover: white space for behaviour-first guardrails under GDPR/MiFID-friendly framing.

Chapter 8. Implementation Plan (0–12 Months)

Milestones:

- Months 0–2: Team, legal perimeter, API landscape scan, design prototype.
- Months 3–4: Clickable prototype; 1–2 broker LOIs; data/DPIA set-up.
- Months 5–6: MVP build; read-only connectors; internal alpha (50 users).
- Months 7–8: Closed beta (500–1,000 users); SDK co-dev with partners.
- Months 9–10: Localisation, impact whitepaper, resilience/security hardening.
- Months 11–12: Public launch in 2–3 EU markets; freemium + Pro; partner co-marketing.

Roadmap visual: